Discrete Math W4

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Hello

I am Diego

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Group Chat



Plan

1. Feedback
2. Theory Recap
3. Exercises
4. Questions?

Feedback

Theory Recap

Remark about predicate logic

When defining a predicate do it like this

$$P(x) = 1 \iff x \geq 1$$

$$P(x) = \begin{cases} 1 & x \geq 1 \\ 0 & \text{else} \end{cases}$$

Also define functions and predicates separately

~~$P(f(y)) \iff f(y) \geq 2$~~

Where Are We?

- Basic Logic & Proofs
- ⇒ Sets & Relations
- Number Theory
- Algebra
- Logic

3. Sets & Relations

- Introduction
- Sets and Operations on Sets
- Relations
- Equivalence Relations
- Partial Order Relations
- Functions
- Countable and Uncountable Sets

The Set Concept

We define an elementhood predicate

$$E(x, y) = 1 \iff x \text{ is element of } y$$

Now we could postulate certain properties on this predicate.

In axiomatic set theory we would define a list of axioms and then build a theory out of them. In this course we use ZFC.

Set Equality

A set is completely specified by its elements.

Definition 3.2 (Axiom of extensionality)

$$A = B \overset{\text{def}}{\iff} \forall x(x \in A \leftrightarrow x \in B)$$

We postulate that if a is a set then $\{a\}$ is also a set.

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$$\{1, 2, 23\} = \{1, 2, 2, 1, 23\} = \{23, 2, 1\} = \{342384728 - 342384727, 2, 23\}$$

Subsets

Definition 3.3

The set A is a subset of the set B, denoted $A \subseteq B$, if every element of A is also an element of B, i.e.,

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Lemma 3.3

For any sets A, B, C.

$$A \subseteq B \land B \subseteq C \implies A \subseteq C$$

Union and Intersection

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The *union* of two sets A and B is defined as

$$A \cup B \stackrel{\text{def}}{=} \{x \mid x \in A \vee x \in B\}$$

and their *intersection* is defined as

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Extension to a Family of Sets

Let $\mathcal{A}$ be a set of sets (a collection).

$$\bigcup \mathcal{A} \stackrel{\text{def}}{=} \{x \mid x \in A \text{ for some } A \in \mathcal{A}\}$$

$$\bigcap \mathcal{A} \stackrel{\text{def}}{=} \{x \mid x \in A \text{ for all } A \in \mathcal{A}\}$$

Some Examples

Let $\mathcal{A} = \{\{1,2\},\{2,3\},\{2,3,4\}\}$

$$\bigcup \mathcal{A} = \{1,2,3,4\} \qquad \text{and} \qquad \bigcap \mathcal{A} = \{2\}$$

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Let $A_n = \{n\}$ for $n \in \mathbb{N}$

$$\bigcup_{n=0}^{\infty} A_n = \mathbb{N} \qquad \text{and} \qquad \bigcap_{n=0}^{\infty} A_n = \emptyset$$

Difference

Definition 3.5

The difference of sets B and A, denoted $B \setminus A$ is the set of elements of B without those that are elements of A

$$B \setminus A \stackrel{\text{def}}{=} \{x \in B \mid x \notin A\}$$

In this course the complement is not defined so don't use it!

Some important laws

Theorem 3.4. *For any sets A, B, and C, the following laws hold:*

Idempotence:	$A \cap A = A$;
	$A \cup A = A$;
Commutativity:	$A \cap B = B \cap A$;
	$A \cup B = B \cup A$;
Associativity:	$A \cap (B \cap C) = (A \cap B) \cap C$;
	$A \cup (B \cup C) = (A \cup B) \cup C$;
Absorption:	$A \cap (A \cup B) = A$;
	$A \cup (A \cap B) = A$;
Distributivity:	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;
Consistency:	$A \subseteq B \iff A \cap B = A \iff A \cup B = B$.

All of these follow directly from the laws about propositional and predicate logic.

The Empty Set

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A set A is called **empty** if it contains no elements. So if

$$\forall x \neg (x \in A)$$

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Lemma 3.6

The empty set is a subset of every set.

$$\forall A (\emptyset \subseteq A)$$

Constructing Sets from the Empty Set

$$\emptyset \neq \{\emptyset\} \neq \{\{\emptyset\}\} \neq \cdots$$

Let

$$A = \{\emptyset\}, \quad B = \{\emptyset, \{\emptyset\}\}, \quad C = \{\emptyset\}, \quad D = \{B, E\}, \quad E = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$$

Find which relations of the form $X = Y$, $X \subseteq Y$, or $X \in Y$ hold between these sets.

Construction of the Natural Numbers

We can define $\mathbb{N}$ from basic set theory using Von Neumann ordinals.[1]

$$\begin{aligned} 0 &= \{\} &&= \emptyset, \\ 1 &= \{0\} &&= \{\emptyset\}, \\ 2 &= \{0, 1\} &&= \{\emptyset, \{\emptyset\}\}, \\ 3 &= \{0, 1, 2\} &&= \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}. \end{aligned}$$

The successor is defined as

$$s(n) \stackrel{\text{def}}{=} n \cup \{n\}$$

Addition can be defined as

$$m + 0 \stackrel{\text{def}}{=} m \quad \text{and} \quad m + s(n) \stackrel{\text{def}}{=} s(m + n)$$

[1] https://en.wikipedia.org/wiki/Set-theoretic_definition_of_natural_numbers

The Power Set

Definition 3.7

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For finite sets $|\mathcal{P}(A)| = 2^{|A|}$

The Cartesian Product of Sets

Definition 3.8

The Cartesian product $A \times B$ of two sets A and B is the set of all ordered pairs with the first component from A and the second component from B

$$A \times B = \{(a, b) \mid a \in A \land b \in B\}$$

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Cartesian product of k sets

$$\bigtimes_{i=1}^{k} A_i = \{(a_1, \ldots, a_k) \mid a_i \in A_i \text{ for } 1 \leq i \leq k\}$$

The Cartesian Product

The cartesian product is not commutative or associative.

Not commutative: for $A = \{1, 2\}$ and $B = \{3\}$

$$A \times B = \{(1, 3), (2, 3)\} \neq \{(3, 1), (3, 2)\} = B \times A$$

Not associative: for $A = \{1\}$, $B = \{2\}$, $C = \{3\}$

$$(A \times B) \times C = \{((1, 2), 3)\} \neq \{(1, (2, 3))\} = A \times (B \times C)$$

since $((1, 2), 3) \neq (1, (2, 3))$.

Further, for any set A

$$A \times \emptyset = \emptyset \times A = \emptyset$$

Exercises

Some exam like tasks

Let $A = \{\emptyset, \{\emptyset\}, 1\}$. How many elements are in $\{A, 3, \emptyset\} \times \{A, A, 1\}$?

Compute the number of elements in the set
$|\mathcal{P}(\{\mathcal{P}(\{0\} \times \emptyset), \{\emptyset\}, 1\} \times \mathcal{P}(\{\emptyset\}))|$

Let $A = \{\{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\{\emptyset\}\}\}\}$. List all subsets of A that are not elements of A.

Let A, B, and C be finite sets such that:

- $|(A \cup B) \times (B \cap C)| = 5$
- $A \cap B = \emptyset$
- $C \subseteq B$ and $|B| = 2$

Find the value of $|A| + |C|$.

Let $|A| = n$.

Compute the number of subsets of $\mathcal{P}(\mathcal{P}(A) \setminus \emptyset) \setminus A$

Let $x \in A$ be fixed. Compute the number of subsets of $\mathcal{P}(A \setminus \{x\})$.

An important set identity

The **symmetric difference** of two sets A and B, denoted by $A \Delta B$, is defined as

$$A \Delta B = (A \setminus B) \cup (B \setminus A)$$

Prove that

$$A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$$

Another Proof

$$A \times (B \setminus C) = (A \times B) \setminus (A \times C)$$